SMART INDIA HACKATHON · INTERNAL HACKATHON

Explainable AI for Diabetic Retinopathy Screening in Rural India

PROBLEM STATEMENT ID
SIH26038

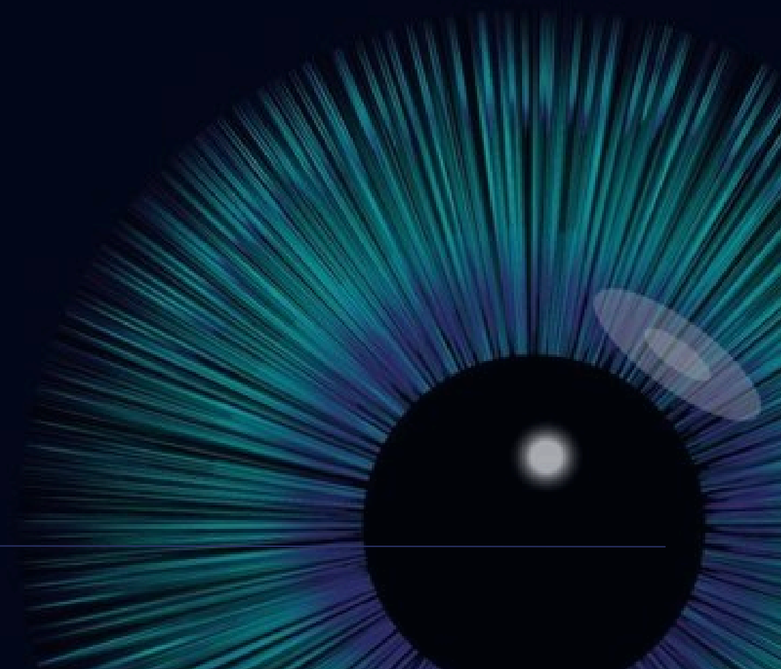
THEME
Clean & Green Technology

CATEGORY
Software

TEAM NAME
Code Catalyst

TEAM LEADER
Jha Naman Chandan

INSTITUTE
IIT Madras BS Degree Programme



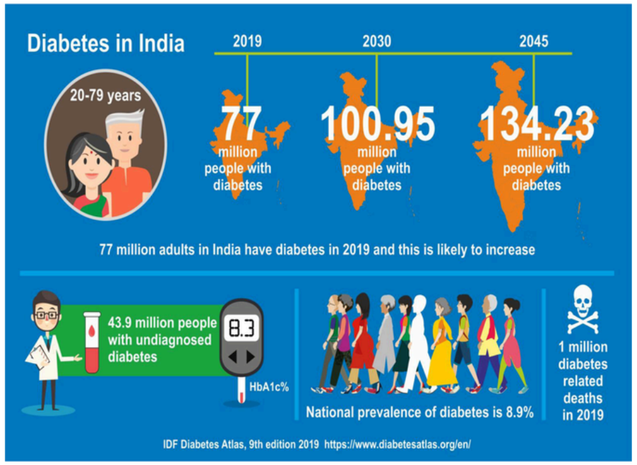
Team Member Details

Member	Name	Role / Contribution
Team Leader	Jha Naman Chandan	*Market Research, Survey + + overall integration*
Member 2	Shreya Sanjay Kachare	*Competitive Analysis, Pitch Deck + Image quality assessment & enhancement*
Member 3	Sandhya Gupta	*Backend, AI ML + Lesion segmentation → microaneurysms, exudates*
Member 4	Vedika Barsaiyan	*Frontend + Explainability → Grad-CAM, calibration*
Member 5	sriram	*DataBase, API + Simulink / SimEvents workflow model*
Member 6	Swatantra Tiwari	*Deployment , Clinical validation protocol, benchmarking & documentation*

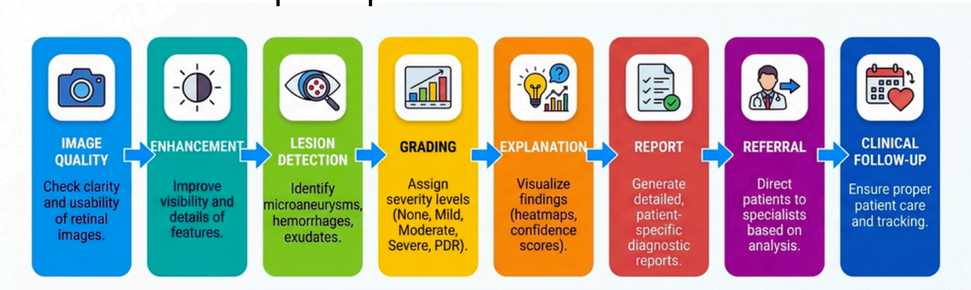
Idea / Approach Detail

Problem

➢ **India's rural eye-care system has a serious shortage, with just 1 ophthalmologist for every 2.5 lakh people. AI-based retinal screening can help, but with 88.43% accuracy and only 68% sensitivity, unclear images and limited trained staff can still result in missed diabetic retinopathy cases.**



➢ Instead of focusing only on DR detection, our system covers the complete process:



Solution

➢ **Better Screening:** Check the quality of retinal images, improve unclear images where possible, and ask the health worker to retake the image if it is too blurry to use.

➢ **Clear & Explainable Detection:** Detect signs of diabetic retinopathy and grade them from ICDR 0–4, while highlighting the affected areas and providing a simple explanation of the result.

➢ **Smart Referral System:** Identify high-risk patients and prioritize them for referral, with offline support so the system can still work in areas with poor internet connectivity.

What Makes It Different

➢ Unlike systems that directly analyse every image, our system first detects blurry/poor-quality images and guides recapture, reducing errors.

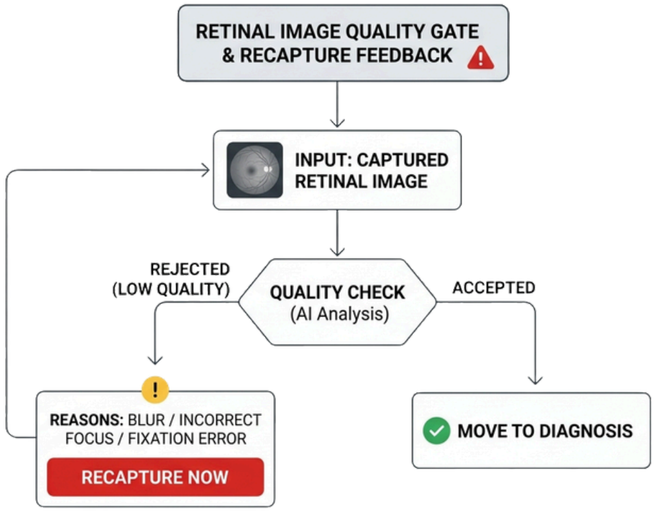
➢ The AI doesn't simply give a DR grade.It highlights the actual lesions responsible for the decision and provides a confidence score, making the result understandable to healthcare workers.

Novelty and Uniqueness

Competitor		Limitation
LumineticsCore® (formerly known as IDx-DR)	→	There is no lesion level explanation. Difficult for opthalmologist to verify the reasoning
EyeArt and Remidio Medios	→	Good accuracy but output is black box i.e it doesn't provide the reasoning behind the grading
Academic Grad-CAM papers	→	Grad-CAM fails on small lesions which is even proven by Saporta/Arun (2021)

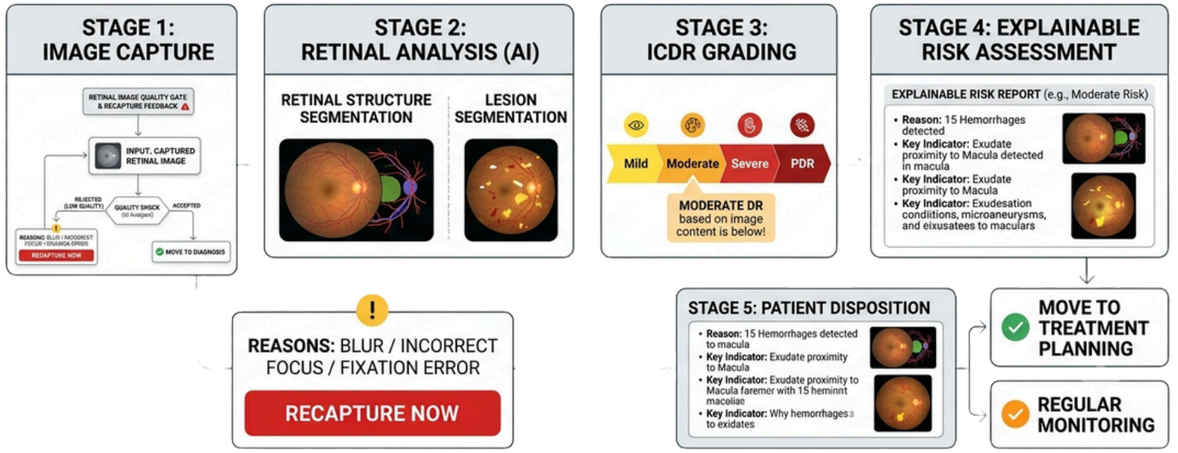
Image Quality Gate + Recapture Feedback :

Before diagnosis, the system automatically checks retinal image quality and guides the operator to recapture poor-quality images, reducing false results.



Complete Multi-Stage AI Pipeline

Instead of only classifying diabetic retinopathy, our system performs retinal structure & lesion segmentation → ICDR grading → explainable risk assessment, showing why a patient was flagged.

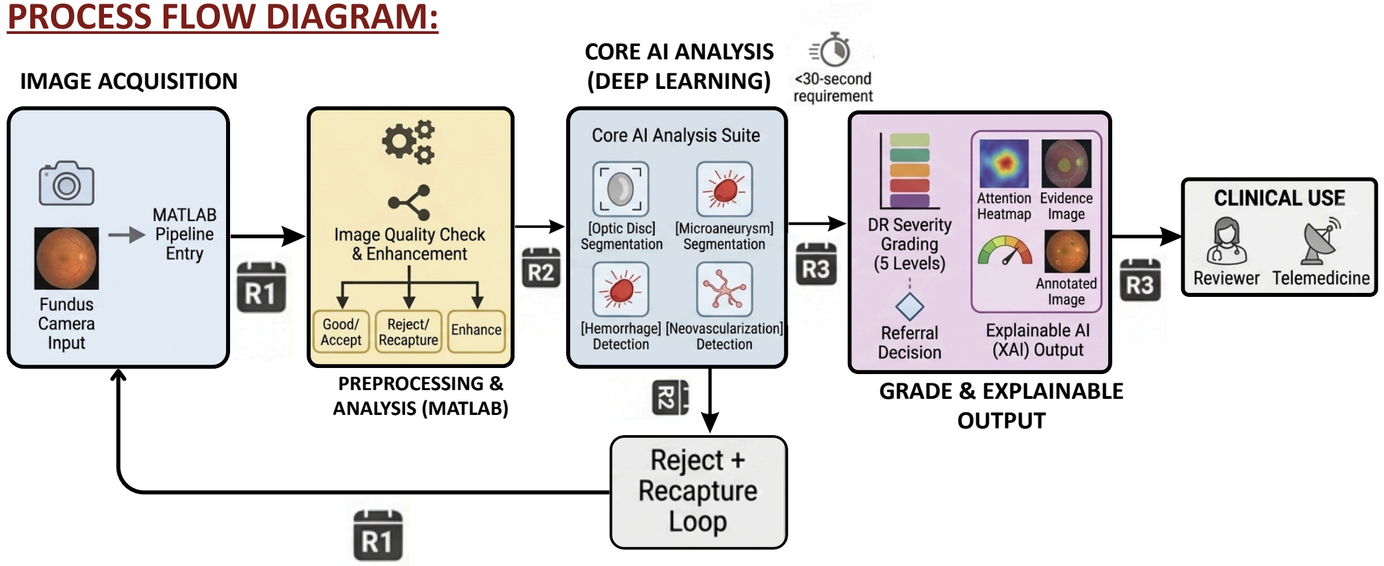


Designed for Rural, Low Resource Deployment

The solution is designed for low-cost fundus cameras, limited internet/connectivity and non-specialist health workers, with an AI-assisted workflow that can support screening and referral at the primary-care level.

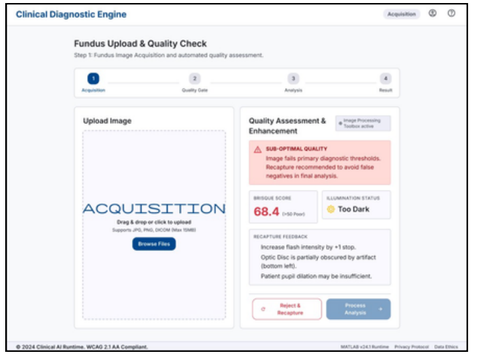
Technical Approach

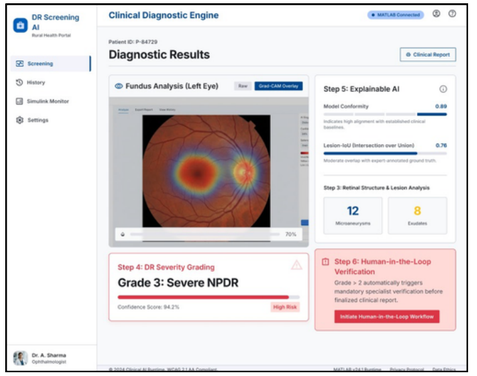
PROCESS FLOW DIAGRAM:

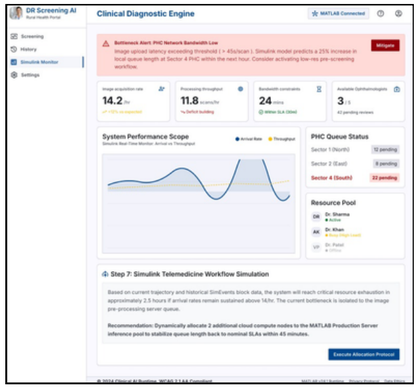


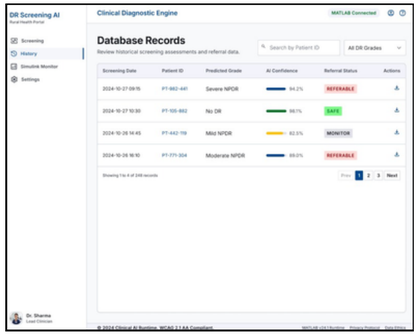
Datasets : **APTOS 2019 (3,662) · IDRiD (516 graded + 81 pixel-annotated) · DRIVE (40 vessels) · Messidor-2 (1,748, external validation)**

Prototype :









TECH STACK:

Frontend: React.js + Bootstrap

Backend: FastAPI + MATLAB Engine API

Database: PostgreSQL

Deployment: Docker + Nginx + MATLAB

AI/ML: MATLAB + Image Processing + Computer Vision + Deep Learning Toolbox

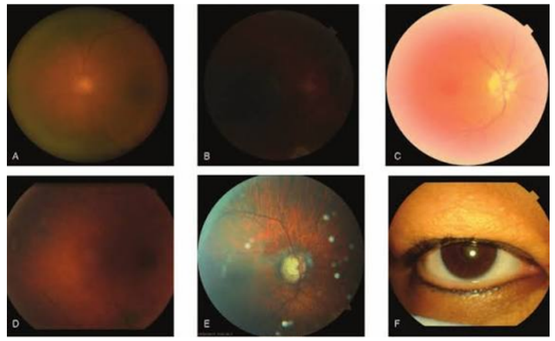
Medical AI: Medical Imaging + Statistics & ML Toolbox + Grad-CAM

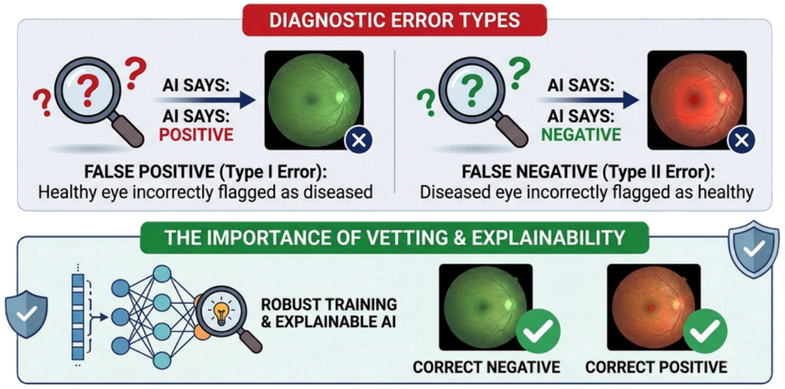
Simulation: Simulink

Feasibility and Viability

Potential Challenges

- Poor-quality / blurry retinal images
- False positives/negatives in AI diagnosis
- Limited connectivity and technical expertise in rural areas





Mitigation Strategy

- Use of an automated image-quality assessment module to detect blur, poor illumination or incomplete field of view and provide recapture feedback before diagnosis.
- Model would be trained and validated on diverse, clinically labelled retinal datasets and threshold based validation would be used with explainable outputs to improve reliability.
- A lightweight, offline and compatible system with a simple interface would be introduced that can be operated by non-specialist healthcare workers, with referral alerts for suspected cases.

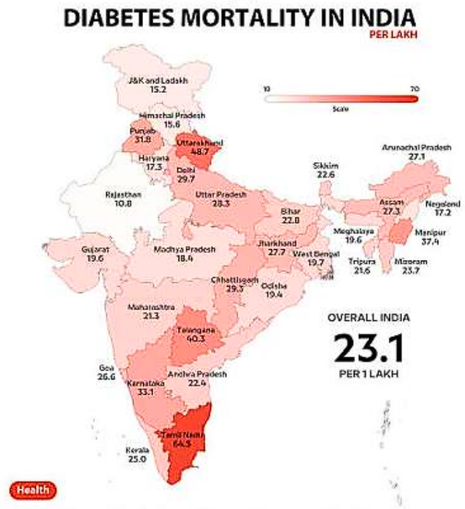
Impact and Benefits

Social Impact

- Approximately 101 million people in India are estimated to suffer from diabetes in 2021.
- 12.5% of diabetic people suffered from diabetic retinopathy, and 4.0% suffered from diabetic retinopathy causing vision loss – around 3 million people above 40 years age across the nation.
- There are 21 million visually impaired and 2.4 million blind people suffering from diabetes.
- Screening for early detection and treatment of diabetic retinopathy can prevent/lessen avoidable visual loss.
- Screening becomes available in rural areas since screening will take place in PHC of villages, thus eliminating repeated visits to the hospital at district level.

Economic Impact

- Lowers specialist workloads – The AI performs triage and sorting of patients, allowing the ophthalmologist's time to be spent on those requiring evaluation or referral.
- Teleophthalmology in rural areas in India has been found to be cost-effective at a rate of $1,320/QALY gained compared to not doing screening; the cost per patient screened by the providers was $7.36.
- The proposed Simulink-based capacity planning model can be used to determine the number of cameras needed, processing capacities and reviewers for a specific number of screenings annually, for example 100,000+.
- Human-in-the-loop screening strikes the balance between efficiency through AI and the involvement of specialists.



Environmental Impact

- Lower movement: tele-screening at the village level will minimize patient travel to district hospital thereby reducing transport-related emissions.
- Paperless processes: digitization of images, artificial intelligence results, and electronic reporting will limit the use of paper work and radiographic films.
- Efficient digital processing: local inference can minimize unnecessary image transmission and thus reduce bandwidth utilization.
- Clean & Green Healthcare: Digital screening helps deliver healthcare with lesser physical inputs and lower movement.



Research and References

*1. Gulshan et al., "**Development and Validation of a Deep Learning Algorithm for Detection of DR in Retinal Fundus Photographs,**" JAMA 2016;316(22):2402-2410*

*2. Gulshan et al., "**Performance of a DL Algorithm vs Manual Grading for DR in India,**" JAMA Ophthalmol 2019*

*3. Porwal et al., "**IDRiD: A Database for DR Screening Research,**" Data 2018;3(3):25 + Porwal et al., "**IDRiD: Segmentation and Grading Challenge,**" Medical Image Analysis 2020*

*4. Selvaraju et al., "**Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization,**" ICCV 2017 / IJCV 2020*

*5. Saporta et al. / Arun et al., "**Deep learning saliency maps do not accurately highlight diagnostically relevant regions,**" (**medRxiv 2021 → Nature Machine Intelligence 2022**)*

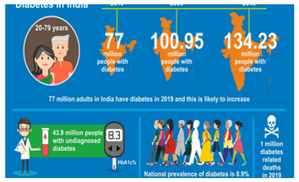
*6. Leibig et al., "**Leveraging uncertainty information from deep neural networks for disease detection,**" Scientific Reports 2017;7:17816*

*7. Rachapelle et al., "**The Cost-Utility of Telemedicine to Screen for Diabetic Retinopathy in India,**" Ophthalmology 2013;120(3):566-573*

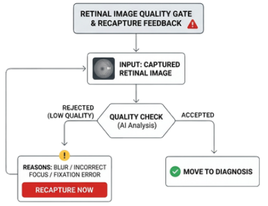
*8. Raman R, Vasconcelos JC, et al., "**Prevalence of DR in India... SMART India study,**" Lancet Global Health 2022;10:e1764-e1773*

Solution Summary

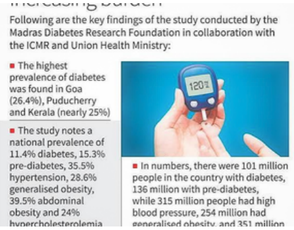
Problem → In India, there are problems with access to DR screening in the rural population, and there are problems with the existing AI algorithms related to poor image quality and clinical explainability.



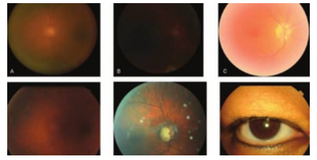
Proposed Solution → Five-step algorithm including an image quality gate, retinal/lesion segmentation, ICDR severity grading, explainable AI, and Simulink-based capacity planning using MATLAB



Novelty / USP → Clinical evidence-based explainability: Grad-CAM is coupled with lesion-level evidence along with overlap/conformity measures, and the ICDR rule layer is used for validating the AI decision.



Feasibility → Depends on the existence of open-source data sets such as APTOS 2019, IDRiD, DRIVE, and Messidor-2, and MATLAB/MathWorks.



Impact → Rural large-scale screening, emphasis on the referable patients who need to visit an ophthalmologist, no burden for the specialists, reduction of patient travel costs.

